

Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

I can only use the prompt and `OnShellBG.m`, and I have to be careful not to read any other pre-existing file. I need a closed-form A_n for $\sigma = (-1, -1, +1, \dots, +1)$, and I should understand how the kernels work for a faster implementation. First question: is `wolframscript` even available? Let's run `wolframscript`. It is. I think about how to use the code: maybe I copy some definitions, maybe I append to `OnShellBG`, maybe I write a separate loader that does `Get`. I write a wrapper that builds the two-minus sign vector, calls `MakeKinematics`, and feeds `BGAmplitude`, then I just `Get` the file — and that was a mistake: `Get` runs the file's own demonstration tests, which are for a *different* sector ($\sigma = (-1, +1, \dots, +1)$, where the amplitude is identically zero), and it prints a wall of “ $A_5 = 0, A_6 = 0, \dots$ ” while the $n = 8$ case grinds. I let it run for a bit, decide it is wasting time, and interrupt it. Better idea: import the file as text, split it at the `VI. TESTS` marker, and `ToExpression` only the part before that, so I load the definitions without the demo. (A `Do[...,{n,fw},...]` trips a “Tag List is Protected” error too, so I switch the loop to `Scan`; small things.)

Now the oracle behaves. The first real surprise is $n = 4$: the recursion returns `Indeterminate` — “ $1/0$ ”, “ $0 \cdot \text{ComplexInfinity}$ ”. The on-shell point forces an internal two-leg current to zero momentum, so a propagator blows up and a real cancellation hides behind a removable $0/0$. I set $n = 4$ aside, to come back as a limit, and look at $n = 5, 6, 7$, which give clean, purely imaginary rationals like $-2304i$. So I write $A_n = i P_n$ and chase P_n .

I keep thinking about kernels, propagators, and the overall dimensions. Scaling $\omega \rightarrow \lambda \omega$ multiplies A_n by λ^{2n-4} ; changing g shows $A_n \propto g^{-(n-3)}$; and a family where the second negative leg goes soft, $\omega_2 = a \rightarrow 0$, shows the amplitude going like a^5 when ω_2 is the smallest scale around. Putting the degree, the g -power, and that soft behaviour together, the simplest thing that fits is the monomial

$$A_n \stackrel{?}{=} i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2^{2n-5}.$$

It is suspiciously clean, and it does reproduce the soft data. But it breaks the instant ω_2 is not the smallest same-sign scale. I notice something intriguing while poking at this: flipping ω_1, ω_2 both negative or both positive gives the same $B \equiv A/(-i)$, so whatever P_n is, it looks symmetric in the two negative legs. That tempts me to fit a symmetric polynomial $f(\omega_1, \omega_2)$ of degree six — $a(u^6 + v^6) + b(u^5v + uv^5) + c(u^4v^2 + u^2v^4) + d u^3v^3$. I solve for the coefficients on four points; the solver returns monstrous fractions, and they miss every other point by huge amounts. So A_n is simply not a polynomial in the two negative frequencies. I toy with a divided-difference / Lagrange form, $\sum_i \sigma_i \omega_i^{2n-4} \prod_{j \neq i} \omega_j / (\omega_i - \omega_j)$, and with B-splines on the sorted (negative) phase velocities, “normal-form coefficients based on the max of the frequencies” — these feel close to the right shape but I have no derivation for them yet.

The derivation is hiding in the absolute values. The only non-analytic thing in the whole construction is $|k_S| = |\sum_{i \in S} \sigma_i \omega_i^2|/g$ inside each propagator. An absolute value has a corner where its argument changes sign, so A_n cannot be one polynomial — it has to be a *spline*, a piecewise polynomial with break surfaces $\sum_{i \in S} \sigma_i \omega_i^2 = 0$. That is exactly why my monomial held in one regime and failed in another: I had been sitting in one cell and then crossed a wall.

So I stop guessing and resolve the absolute values by hand, chamber by chamber. I take the symbolic $n = 5$ amplitude, collect every `Abs` argument, replace each one by $\pm$ itself according to its sign at a sample point in the cell, and simplify. Inside each cell the expression is an honest rational function, and after dividing

out the common prefactor $i 2^4 \omega_1 \omega_2$ the pieces are, with $x = \omega_2$ and the positive legs y, z ,

$$x < y, z: x^4, \quad y < x < z: x^4 - (x^2 - y^2)^2, \quad y, z < x: x^4 - (x^2 - y^2)^2 - (x^2 - z^2)^2 + (x^2 - y^2 - z^2)^2.$$

These are not separate answers; they are the running inclusion–exclusion corrections of one object. The function “looks like $x^4 - \sum (x^2 - y_j^2)_+^2$ ”. Writing $t = \omega_2^2$, $a_j = \omega_j^2$, $m = n - 3$, and using the truncated power $[u]_+ = \max(u, 0)$, every cell is one value of

$$T_m(t; \{a_j\}_{j \in R}) = \sum_{S \subseteq R} (-1)^{|S|} \left[t - \sum_{j \in S} a_j \right]_+^m,$$

the truncated-power / B-spline form I had been circling. The walls $t = \sum_{j \in S} a_j$ are the $|k_S|$ sign changes, and below all of them only the $S = \emptyset$ term lives, so $T_m = t^m$ — which is precisely the principal-chamber monomial I started from. The whole thing is one truncated power times a universal prefactor,

$$A_n = i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2 T_{n-3}(\omega_2^2; \{\omega_j^2\}_{j=3}^{n-1})$$

with $R = \{3, \dots, n-1\}$ the $n-3$ positive plus-legs.

Two loose ends. For $n = 4$ the spline has $m = 1$, so the formula predicts $A_4 = i(8/g) \omega_1 \omega_2 \min(\omega_2^2, \omega_3^2)$, and I recover the removable $0/0$ honestly as the $\varepsilon \rightarrow 0^+$ limit of the energy-conserving deformation $\omega = \{-\omega_3 - \varepsilon, \omega_2, \omega_3, -\omega_2 + \varepsilon\}$; the limit is finite and gives $-192i$ for $\{2, 3\}$, matching. The other loose end is which leg is “ t ”, and here I genuinely stumble: my first channel guess predicts -3776 where the oracle says -701 for one point, and it is fine for a small ω_2 but goes wrong once ω_2 is not small. I think the discrepancy is about signs, and I reframe it as a channel-selection question: counting left- and right-movers, my form works for one negative leg against $n-3$ same-sign plus legs, and the right rule is to let p be the $\sigma = -$ leg whose sign is shared by exactly $n-3$ of the plus legs (the set R), with q the other $\sigma = -$ leg, so $A_n = i 2^{n-1} q p T_{n-3}(p^2; \{\omega_j^2\}_{j \in R}) / g^{n-3}$.

My first script for this channel-aware form fails on every line — but the failure is a **Function::slotn** slot mistake in my selector code, not a disagreement with the oracle. I fix the selector and rerun, and now the closed form reproduces **BGAmplitude** *exactly*: the relative error is identically zero in exact rational arithmetic for $n = 4, 5, 6, 7$, at many points per n , including non-generic regimes where one frequency is a hundred times larger or smaller than the others, and at $g \neq 1$.

I want to be honest about the scope rather than overclaim. For arbitrary positive free frequencies handed to **MakeKinematics** — the setup the prompt actually specifies — energy conservation drives the two solved legs negative and the kinematics always land in this one physical channel; I check five hundred random points for each $n = 4, \dots, 8$ and the boxed formula is exact on all of them. If instead one hand-flips the signs of the free frequencies into other cells of the spline, the same truncated-power structure persists but with a different leg as p and a different same-sign set as R ; I state that selection rule but I did not fold every sign cell into a single universal expression. So, honestly: within the prompt’s own kinematics this is the closed-form answer and it is verified to exact equality; the fully sign-general statement I leave as the channel-selection rule plus the spline structure, not as one compressed formula.